

1. Apply all the CPU scheduling algorithms for insem examination of OSCP

	AT	BT	Priority
P ₁	0	4	3
P ₂	0	9	1
P ₃	0	6	2
P ₄	0	4	3
P ₅	0	9	1
P ₆	0	6	2
P ₇	1	4	3
P ₈	1	9	1
P ₉	1	6	2

2. Define following terms

- * Starvation
- * Convoy effect
- * Context Switching
- * Aging

3. A Mobile has the following tasks, draw a table with

Task	AT	BT
Music Player	0	20
Touch even	3	1
Whatsapp	5	2
Camera	6	4
Launch		

- (i) Which Scheduling algorithm is best suitable for above problem
 (ii) Draw the Gantt chart for SRTF
 (iii) Why selected algorithm for improve user experience
 (iv) Which task receives the fastest response

4. Solve the following problem with round robin algorithm time quantum of 0.5 Draw a table

Process	AT	BT
P ₁	0	5
P ₂	0	4
P ₃	0	2

Solutions

1. FCFS - First Come First Serve

Ready Queue	P ₁	P ₂	P ₃	P ₄	P ₅	P ₆	P ₇	P ₈	P ₉
Arrival time	0	0	0	0	0	0	1	1	1

Gantt Chart	P ₁	P ₂	P ₃	P ₄	P ₅	P ₆	P ₇	P ₈	P ₉
	0	4	13	19	23	32	38	42	51

$$TAT = CT - AT$$

$$WT = TAT - BT$$

$$= CT - AT - BT$$

P₁

P₂

P₃

P₄

P₅

P₆

P₇

P₈

P₉

Avg TAT

Avg WT

SJN (Shortest Job First)

Process

P₁

P₂

P₃

P₄

P₅

P₆

Smallest BT

Gantt Chart

P ₁	P ₂
0	4

	AT	BT	CT	TAT	WT
P ₁	0	4	4	4	0
P ₂	0	9	13	13	4
P ₃	0	6	19	19	13
P ₄	0	4	23	22	19
P ₅	0	9	32	32	23
P ₆	0	6	38	38	32
P ₇	1	4	42	41	37
P ₈	1	9	51	50	41
P ₉	1	6	57	56	50
				<u>276</u>	<u>219</u>

$$\text{Avg TAT} = \frac{276}{9} = 30.667$$

$$\text{Avg WT} = \frac{219}{9} = 24.333$$

SJN (Shortest Job Next) / shortest Job first (SJF) :-

Process	AT	BT	Process	AT	BT
P ₁	0	4	P ₇	1	4
P ₂	0	9	P ₈	1	9
P ₃	0	6	P ₉	1	6
P ₄	0	4			
P ₅	0	9			
P ₆	0	6			

Smallest BT = 4

Gantt Chart

P_1	P_4	P_7	P_3	P_6	P_9	P_2	P_5	P_8	
0	4	8	12	18	24	30	39	48	57

	AT	BT	CT	TAT	WT
P ₁	0	4	4	4	0
P ₄	0	4	8	8	4
P ₇	1	4	12	11	7
P ₃	0	6	18	18	12
P ₆	0	6	24	24	18
P ₉	1	6	30	29	23
P ₂	0	9	39	39	30
P ₅	0	9	48	48	39
P ₈	1	9	57	56	47
			<u>237</u>	<u>180</u>	

$$\text{Avg TAT} = \frac{237}{9} = 26.33$$

$$\text{Avg WT} = \frac{180}{9} = 20$$

SRTF (shortest Remaining Time first)

At Time 0 P₁ starts RBT = 4 - 1 = 3
at time 1 P₁, P₈, P₉ arrive.

P ₁	3
P ₄	4
P ₇	4
P ₃	6
P ₆	6
P ₉	6
P ₂	9
P ₅	9
P ₈	9

Grantt

Priority

Higher

P₉ →

0 → 9

9 → 18

18 → 27

Priority

27 → 33

33 → 39

39 → 45

Grantt Ck

CT

P₁ 49

P₂ 9

P₃ 33

P₄ 53

P₅ 18

P₆ 39

P₇ 57

P₈ 27

P₉ 45

Grantt Chart

P_1	P_4	P_7	P_3	P_6	P_9	P_2	P_5	P_8	
0	4	8	12	18	24	30	39	48	57

Same values as SJN.

Priority Scheduling - (Non Preemptive)

Highest Priority = 1

P₉ → P₅

0 → 9 : P₂

9 → 18 : P₅

18 → 27 : P₈

Priority 2

27 → 33 : P₃

33 → 39 : P₆

39 → 45 : P₉

Priority 3:-

45 → 49 : P₁

49 → 53 : P₄

53 → 57 : P₇

Grantt Chart

	CT	TAT	WT
P ₁	49	49	45
P ₂	9	9	0
P ₃	33	33	27
P ₄	53	53	49
P ₅	18	18	9
P ₆	39	39	33
P ₇	57	57	52
P ₈	27	27	17
P ₉	45	45	38
	<u>327</u>	<u>327</u>	<u>270</u>

$$\text{Avg TAT} = \frac{327}{9} = 36.33$$

$$\text{Avg WT} = \frac{270}{9} = 30$$

Priority Scheduling (Preemptive)

At time 0, P_2 has Priority 1

$0 \rightarrow 1 : P_2$

$0 \rightarrow 9 : P_2$

$9 \rightarrow 18 : P_5$

$18 \rightarrow 27 : P_8$

Priority 2 :

$27 \rightarrow 33 : P_3$

$33 \rightarrow 39 : P_6$

$39 \rightarrow 45 : P_9$

So, for this particular data

Preemptive = Non Preemptive

Priority 3 :-

$45 \rightarrow 49 : P_1$

$49 \rightarrow 53 : P_4$

$53 \rightarrow 57 : P_7$

2. (i) Starvation :-

Occurs when a process waits indefinitely or for an excessively long time because other processes are repeatedly selected for execution. Starvation can occur in both preemptive and Non-preemptive scheduling, depending on the algorithm.

(ii) Convoy effect :-

is an important problem associated with non-preemptive scheduling especially FCFS. The long process acts like a convoy, causing many short processes to follow behind it.

(iii) Con

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process
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(iv) Agin

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3.

(i) SRTF
with th

(ii) At +
Music

RBT =
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To

0 3
Music | T

(iii) Context Switching :-

is the process of saving the state of the currently running process and loading the saved state of another process so that the CPU can switch between them.

(iv) Aging :-

is a technique used to prevent starvation by gradually increasing the priority of a process that has been waiting for a long time.

3.

(i) SRTF always selects the process with the shortest remaining Burst time

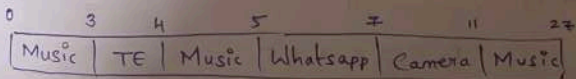
(ii) At time 0: only Music Player.

Music player has executed for 3 units

$$RBT = 20 - 3 = 17$$

At time 3: Touch Event arrives

	RBT
Music player	17
Touch Event	1



(iii) SRTF improves the user experience because it gives short interactive tasks priority over long CPU-intensive tasks

(iv) $RT = \text{First CPU start time} - AT$

	AT	FS	time - AT
Music	0	0	0
Touch	3	3	0
Whatsapp	5	5	0
Camera	6	7	1

The fastest Response time are:
Music Player, Touch Event,
Whatsapp Notification.

4.

P ₁	P ₂	P ₃
0	0	0

* P₁ requires 5ms

P₁ : 0 → 2

Remaining : 5 - 2 = 3ms

Context Switch = 2 → 2.5

P₂ → P₃ → P₁

* P₂ requires 4

P₂ : 2.5 → 4.5

Remaining

4 - 2 = 2

P₃ → P₁ → P₂

Context S

* P₃ req

P₃

P₁ →

* P₁ has 3

P₁ :

3 - 2 =

9.5 → 10

P₂ → P₃

0	2	2.5
P ₁	CS	P ₂

P ₃	CS	P ₁
13.5	14	15

P₁

P₂

P₃

TAT = CT

P₁ TAT 15

P₂ 12

P₃ 13.5

Avg waiting

Avg TAT =

experience
tive

Context Switch: $4.5 \rightarrow 5$

* P_3 requires 3ms

$$P_3: 5 \rightarrow 7$$

$$3 - 2 = 1$$

$$P_1 \rightarrow P_2 \rightarrow P_3$$

* P_1 has 3ms remaining

$$P_1: 7.5 \rightarrow 9.5$$

$$3 - 2 = 1$$

$$9.5 \rightarrow 10$$

$$P_2 \rightarrow P_3 \rightarrow P_1$$

0	2	2.5	4.5	5	7	7.5	9.5	10	12	12.5
P_1	CS	P_2	CS	P_3	CS	P_1	CS	P_2	CS	
P_3	CS	P_1								
13.5	14	15								

CT

$$P_1 \quad 15$$

$$P_2 \quad 12$$

$$P_3 \quad 13.5$$

$$TAT = CT - AT$$

	TAT	WT
P_1	15	10
P_2	12	8
P_3	13.5	10.5

Avg waiting time = 9.5

$$\text{Avg TAT} = \frac{15 + 12 + 13.5}{3} = \frac{40.5}{3}$$